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(b) The average kinetic energy per molecule is $\frac{3}{2}kT$. It depends only on absolute temperature of the gas and is independent of the nature of molecules. Since both the gas molecules are at the same temperature, they have the same average kinetic energy per molecule. k is Boltzmaan constant.

$$\frac{3}{2}kT = \frac{3}{2}\times 1.38\times 10^{-23}\times 300 = 6.21\times 10^{-21} J$$

(c) Average kinetic energy of total oxygen molecules = $\frac{3}{2}N_O kT$ where N_o- number of oxygen molecules in the room

Average kinetic energy of total hydrogen molecules = $\frac{3}{2}N_H kT$ where N_H- number of hydrogen molecules in the room.

It is given that the number of oxygen molecules is 3 times more than number of hydrogen molecules in the room. So the ratio of average kinetic energy of oxygen molecules with average kinetic energy of hydrogen molecules is 3:1

9.2.6 Mean (or) average speed ($\bar{v}$)

It is defined as the mean (or) average of all the speeds of molecules

If v_1, v_2, v_3.....v_N are the individual speeds of molecules then

$$\bar{v} = \frac{v_1 + v_2 + v_3 ..... + v_n}{N} = \sqrt{\frac{8RT}{\pi M}} = \sqrt{\frac{8kT}{\pi m}} \quad (9.20)$$

Here M- Molar Mass and m – mass of the molecule.

$$\bar{v} = 1.60\sqrt{\frac{kT}{m}} \quad (9.21)$$

9.2.7. Most probable speed (V_{mp})

It is defined as the speed acquired by most of the molecules of the gas.

$$v_{mp} = \sqrt{\frac{2RT}{M}} = \sqrt{\frac{2kT}{m}} \quad (9.22)$$

$$v_{mp} = 1.41\sqrt{\frac{kT}{m}} \quad (9.23)$$

The derivation of equations (9.20), (9.22) is beyond the scope of the book

Comparison of v_{rms}, $\bar{v}$ and v_{mp}

Among the speeds v_{rms} is the largest and v_{mp} is the least

$$v_{rms} > \bar{v} > v_{mp}$$

Ratio-wise,

$$v_{rms} : v : v_{mp} = \sqrt{3} : \sqrt{8/\pi} : \sqrt{2} = 1.732 : 1.6 : 1.414$$

EXAMPLE 9.3

Ten particles are moving at the speed of 2, 3, 4, 5, 5, 5, 6, 6, 7 and 9 m s^{-1}. Calculate rms speed, average speed and most probable speed.

Solution

The average speed

$$\bar{v} = \frac{2+3+4+5+5+5+6+6+7+9}{10} = 5.2\, m\, s^{-1}$$

To find the rms speed, first calculate the mean square speed $\overline{v^2}$

$$\overline{v^2} = \frac{2^2+3^2+4^2+5^2+5^2+5^2+6^2+6^2+7^2+9^2}{10}$$

$$= 30.6\, m^2\, s^{-2}$$

The rms speed

$$v_{rms} = \sqrt{\overline{v^2}} = \sqrt{30.6} = 5.53\, m\, s^{-1}$$

The most probable speed is 5 m s^{-1} because three of the particles have that speed.